



Test Report No. 2018 – 0479



Agricultural Machinery – Forage Chopper EXCELHIGH FC-2A

Test Report Number	2018 – 0479
Valid Until	July 13, 2023
Agricultural Machinery	Forage Chopper
Type	Random-cut, Flywheel cutterhead, Throw-away
Brand	EXCELHIGH
Model	FC-2A
Test Requested by	WINCH MECHANICAL INNOVATIONS & GENERAL CONSTRUCTION, INC. Airport Road, National Highway, Brgy. Fatima, General Santos City

SUMMARY

Performance Requirements	Requirement as per PAES 218:2004	AMTEC Test
Variation of cut, %, maximum	Not specified by the manufacturer	42.7
Quality of chopped material	Clear-cut	Clear-cut
Noise level, dB(A), maximum	96.0*	103.4

Note: *Allowable noise level for four (4) hours of continuous exposure based on Occupational Safety and Health Standards, Ministry of Labor, Philippines, 1983

OBSERVATIONS

1. Operator's manual	Not Provided
2. Previous valid test reports of similar brand and model	None
3. Prime mover	7.46 kW YAMADA 186F Diesel Engine
4. Manufacturer	EXCELHIGH INDUSTRIAL TRADING AND SERVICES Brgy. Mabuhay, General Santos City
5. Loading	The test materials were directly fed to the machine.
6. Cleaning of parts	Parts were accessible for cleaning.
7. Adjusting and repair of parts	Parts were accessible for adjustments and repairs.
8. Collecting of output	The output fell from the chute and collected directly from the ground after the operation.
9. Transporting the machine	The machine was carried manually.
10. Safety	Chopping assembly was covered.
11. Number of operators	Two (one operator and one bagger)
12. Other remarks	No breakdown during test.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	June 8, 2018
Location of Test	Brgy. Mabuhay, General Santos City
Items Tested	Noise Level, Fuel Consumption, Input Capacity, Chopping Efficiency, Coefficient of Variation, and Number of Operatos

METHODS OF TEST

The forage chopper was tested based on PNS/PAES 219:2004 – Agricultural Machinery – Forage Chopper – Methods of Test.

DESCRIPTION OF THE MACHINE

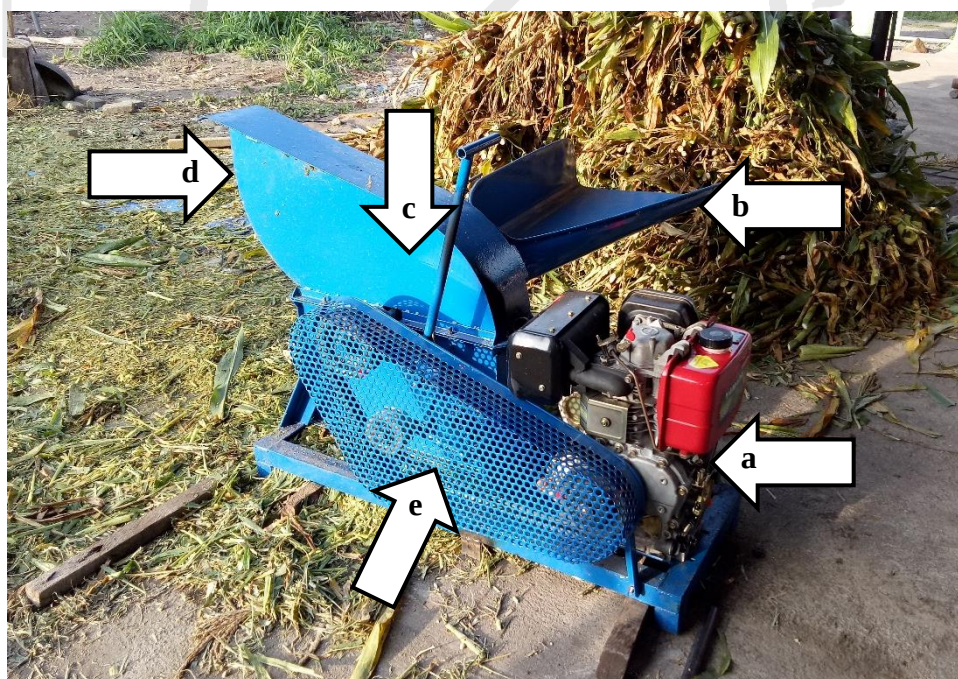


Figure 1. Parts of EXCELHIGH FC-2A Forage Chopper powered by 7.46 kW YAMADA 186F Diesel Engine:
 (a) Prime mover, (b) Feeding table, (c) Chopping chamber, (d) Output chute, and
 (e) Belt and pulley transmission system with cover.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification ^b
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	ND	1365
1.1.2	Width	ND	1350
1.1.3	Height	ND	870
1.2	Weight without prime mover, kg	ND	NM
2	Chopping assembly		
2.1	Type	ND	Random-cut
2.2	Size (D), mm	ND	515
2.3	Cutting device		
2.3.1	Type	ND	Flywheel cutterhead
2.3.2	Dimension (L x W x T), mm	ND	195 x 58 x 7
2.3.3	Number of knives	ND	3
2.3.4	Means of attachment	ND	Bolted
2.3.5	Materials	ND	Tool steel
3	Feeding table		
3.1	Dimension, (L x W), mm	ND	990 x 500
3.2	Height from the ground, mm	ND	520
3.3	Material	ND	3 mm G.I. sheet

Note: ND – No Data, NM – Not Measured

^aThe requesting party did not submit machine specifications.

^bData obtained from sticker, nameplate, actual test, and on-site measurements.

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification ^b
4	Feeding inlet dimension (W x H), mm	ND	160 x 140
5	Output chute dimension (W x H), mm	ND	135 x 335
5.1	Height from the ground, mm	ND	485
6	Chopping chamber		
6.1	Dimension (D), mm	ND	600
6.2	Material of construction	ND	ND
7	Prime mover		
7.1	Brand	ND	YAMADA
7.2	Model	ND	186F
7.3	Serial number	ND	ND
7.4	Type	ND	4-stroke cycle, diesel engine
7.5	Rated power, kW	ND	7.46
7.6	Rated speed, rpm	ND	ND
7.7	Cooling system	ND	Air-cooled
7.8	Starting system	ND	Rope recoil starting
7.9	Weight, kg	ND	NM
8	Transmission system	ND	V-belt and pulley
8.1	Engine pulley	ND	152 mm x 2B x NM
8.2	Chopper pulley	ND	152 mm x 2B x 35 mm

Note: ND – No Data, NM – Not Measured

^aThe requesting party did not submit machine specifications.

^bData obtained from sticker, nameplate, actual test, and on-site measurements.

RESULTS

Item		Data	
1	Conditions of test sample		
1.1	Test sample	Corn stalks	
1.2	Source	Polomolok, South Cotabato	
1.3	Moisture content, %	82.4	
1.4	Dimensions (L x D), mm	1341 x 25	
2	Performance test		
2.1	Operating time, h	0.279	
2.2	Weight of input, kg	524	
2.3	Input capacity, kg/h	1878	
2.4	Chopping efficiency, %	100.0	
2.5	Speed of components, rpm	Without load	With load
2.5.1	Prime mover	2376	2283
2.5.2	Chopping chamber shaft	2104	1905
2.6	Noise level, dB(A)	102.6	103.4
2.7	Fuel consumed, mL	495	
2.8	Fuel consumption, L/h	1.48	
2.9	Number of operators	Two (one operator and one bagger)	
2.10	Average length of cut, mm	39.0	
2.10.1	Coefficient of variation, %	42.7	

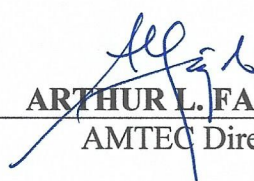
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AUG 19 2019
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